APPLIED MACHINE LEARNING AND ECONOMETRICS CASE

Korea Income & Welfare

This project studies how education relates to income in Korean welfare microdata. It now works as a stronger portfolio case because the modeling is broader, the best predictive model is tuned, and the final conclusion is easier to explain to a non-technical audience.

89,935

Valid records used in the analytical layer

0.645

Best out-of-sample R-squared from the tuned random forest

12.47%

Average income premium per education step in the OLS model

What question does it answer?

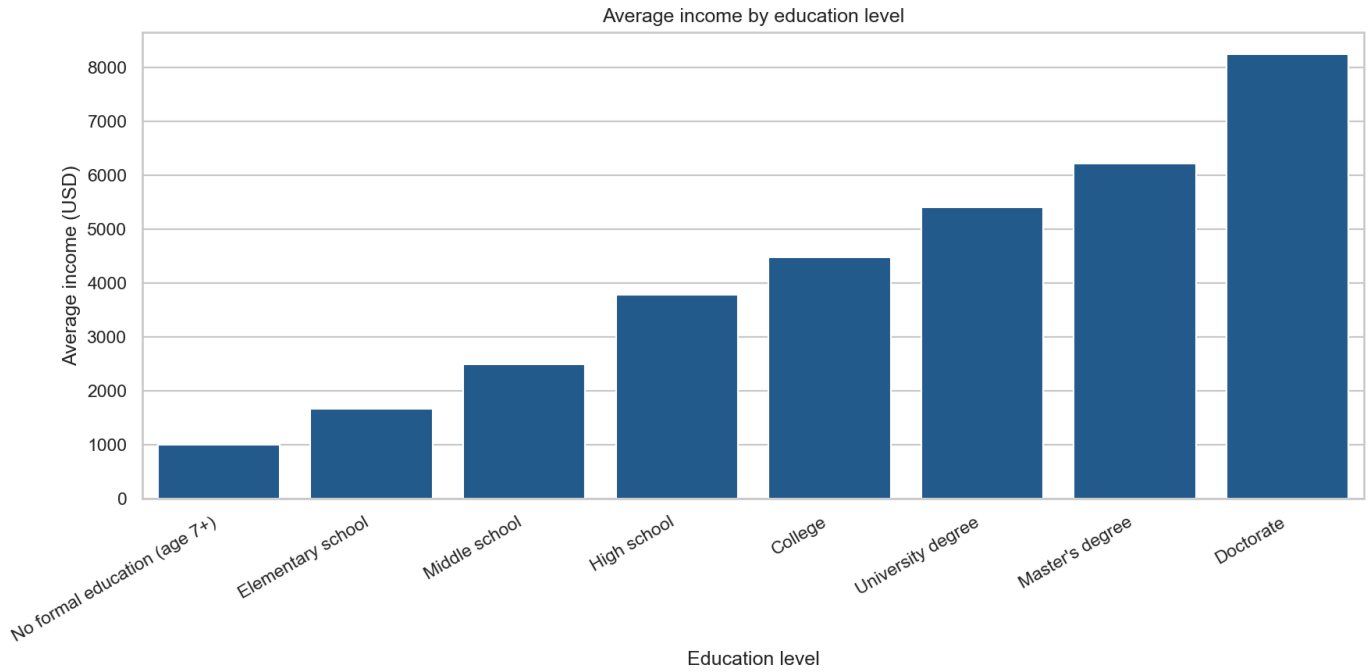
The case asks three practical questions. First, how much can education explain about income? Second, how much predictive power appears once age, household structure, region, and demographics are included? Third, does education pay off equally across the income distribution?

> Executive takeaway: education clearly matters, but household structure and life-cycle effects matter too, and the payoff to education becomes stronger in the upper part of the income distribution.

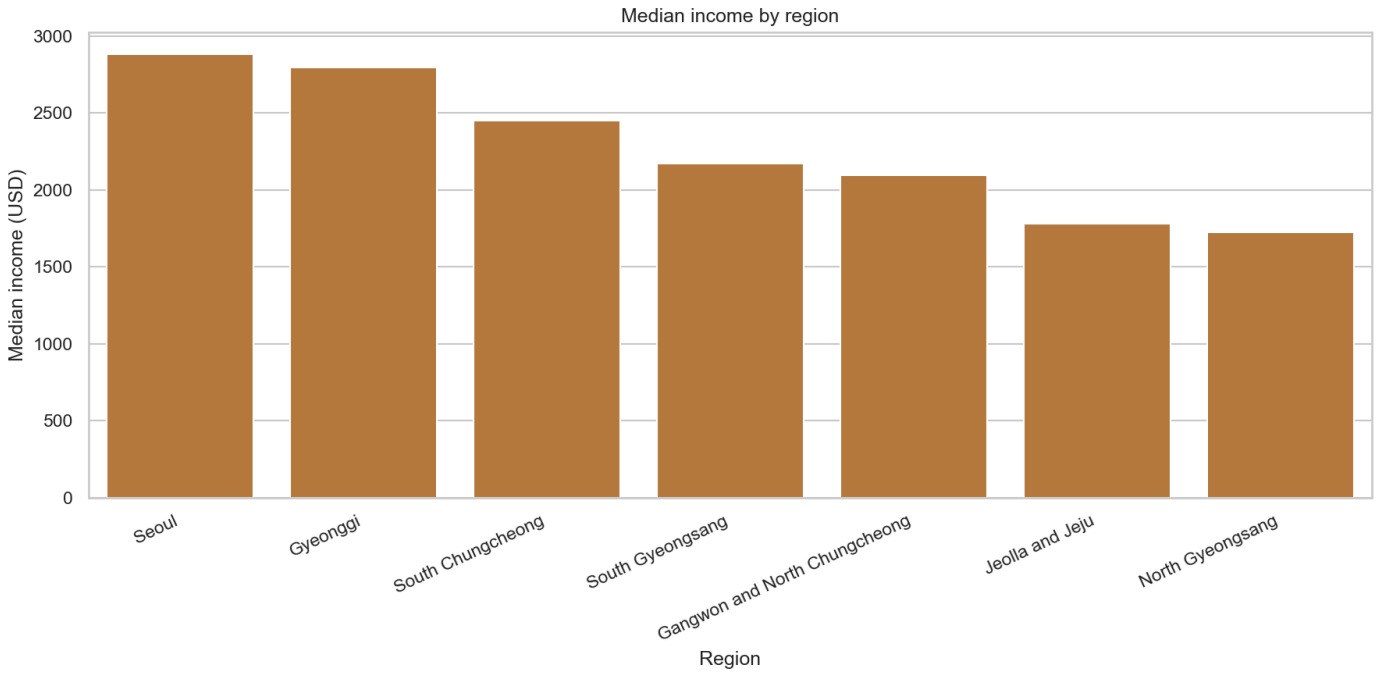
- The tuned random forest improves the original predictive benchmark from 0.294 to 0.645 in R-squared.
- The best model uses 400 trees, max depth 16, and minimum leaf size 3.
- Quantile regression shows the education premium rising from 11.46% to 14.17% as income increases.

What the descriptive layer shows

Before any model is trained, the data already show a clear socioeconomic pattern: income rises with education, and Seoul leads the regional distribution.



Income increases steadily across the education ladder, which makes the central hypothesis intuitive even for a non-technical reader.

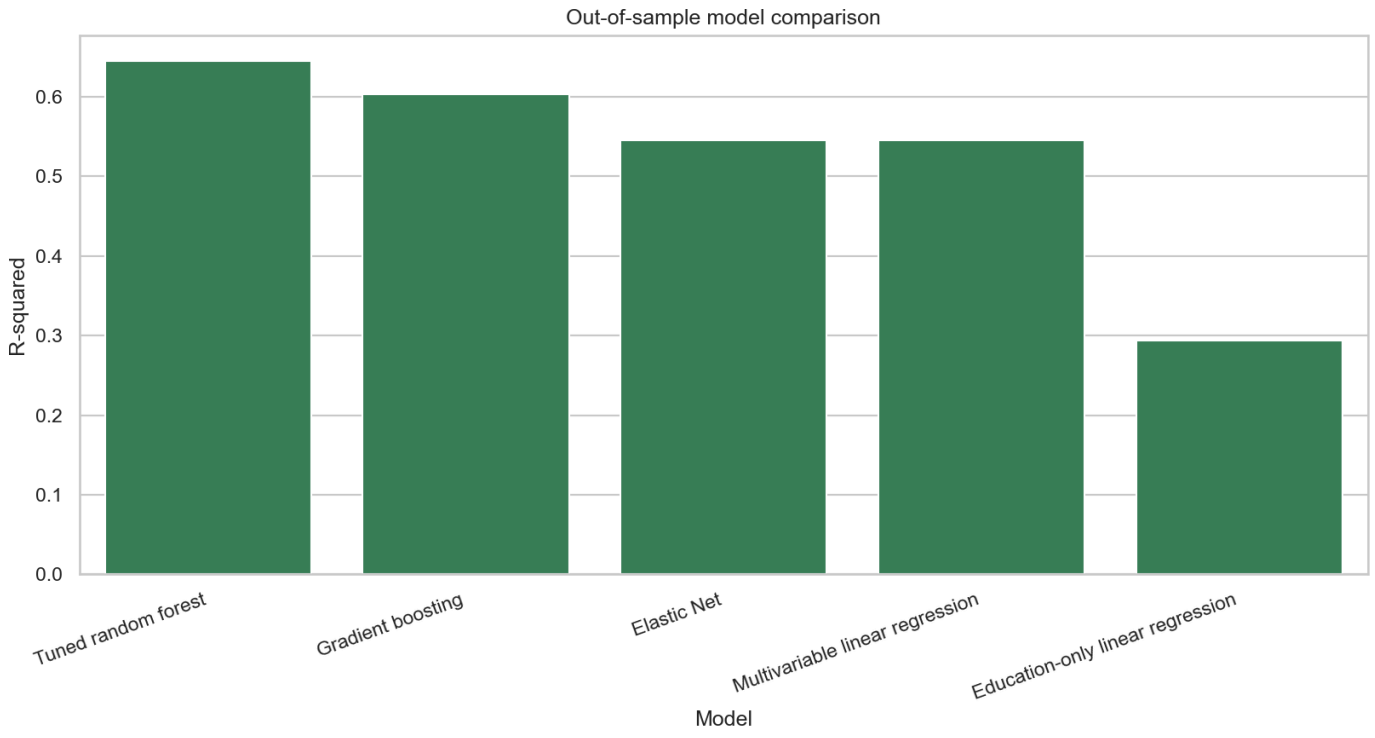


Regional context matters. Education is important, but it is not the only part of the income story.

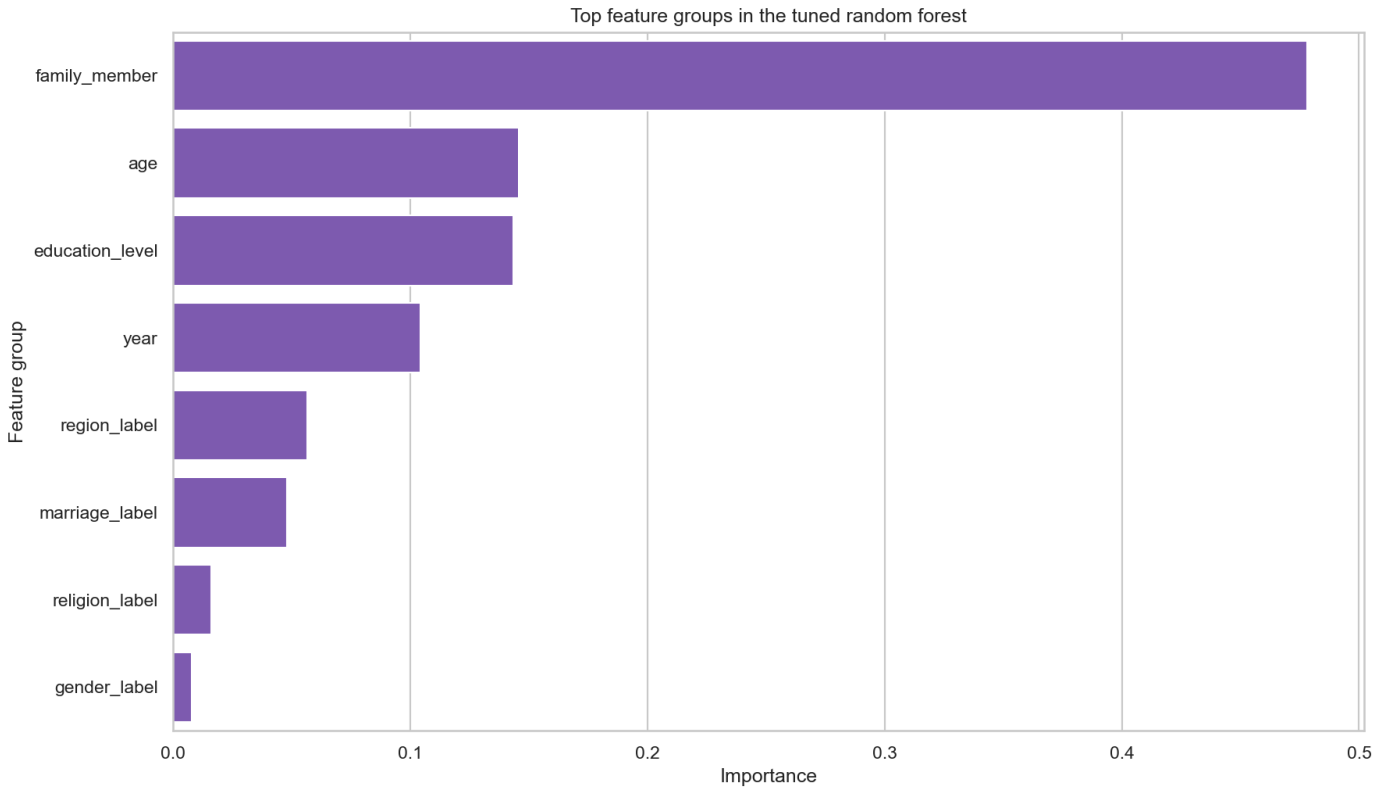
What the modeling layer adds

The repository now compares five out-of-sample predictive models, then complements that with coefficient-based econometrics. This combination is valuable because it shows both performance and interpretation.

Model	MAE (USD)	RMSE (USD)	R-squared
Tuned Random Forest	880.05	1,249.11	0.645
Gradient Boosting	950.31	1,321.48	0.603
Elastic Net	1,049.76	1,414.42	0.545
Multivariable Linear Regression	1,049.83	1,414.49	0.545
Education-only Linear Regression	1,345.40	1,762.56	0.294



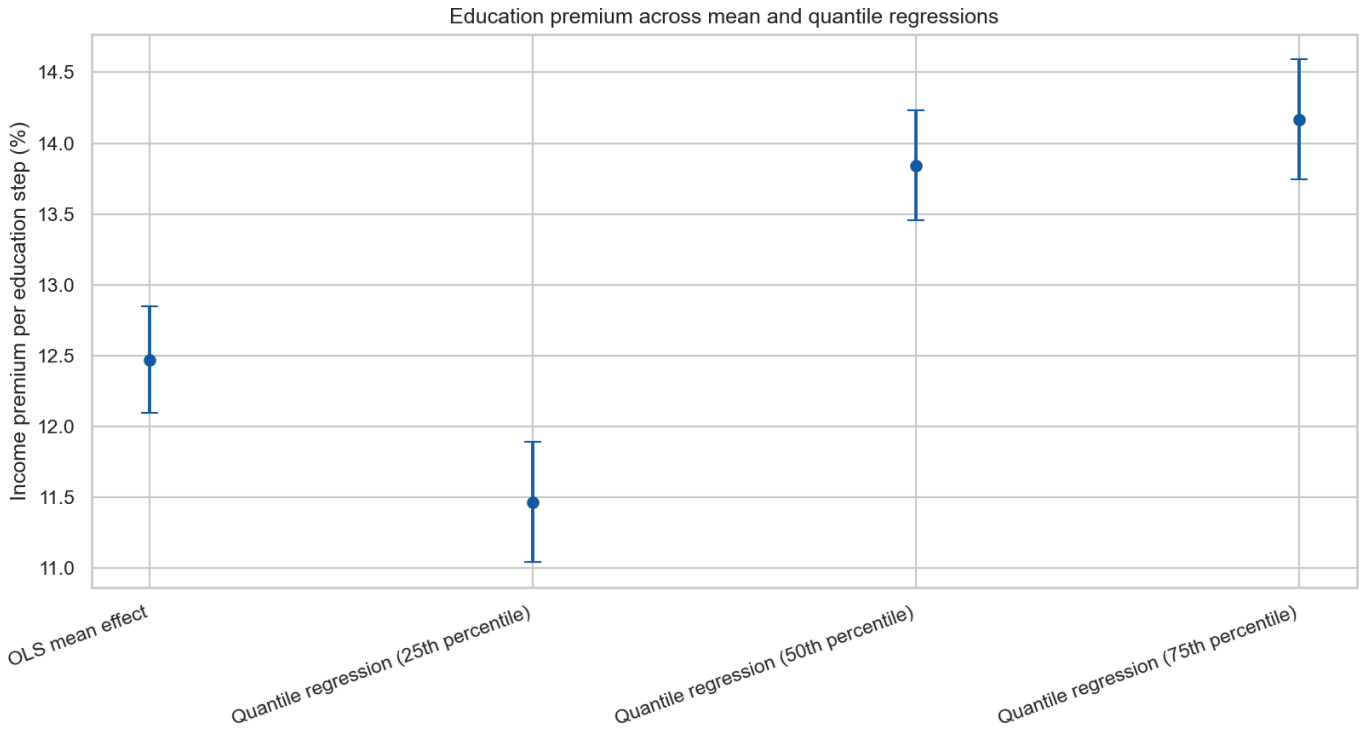
The predictive story is much stronger after adding multivariable structure and tuning. The tuned random forest becomes the clear winner.



Household size is the most important feature group in the best model, followed by age, education, and survey year.

What the econometrics say

The econometric layer gives the project a more interview-friendly conclusion. Instead of only saying that education predicts income, it shows that the education premium grows across the distribution.



The estimated premium is 12.47% on average, 11.46% near the lower end of the distribution, and 14.17% in the upper part. That makes the story richer than a single mean effect.

Hiring takeaway: this project is strong because it combines tuning, benchmarking, and interpretation. It shows that model performance improved, but it also explains what changed in economic terms.